

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 02

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Recap of Previous Lecture



Topic

Architecture vs Organization

Topic

Numbers & Data in Computers

Topic

Components of Computer

CPU, I/O, mem.

Topic

System Buses

Topic

Types of Buses

Topics to be Covered



Topic

CPU Registers

Topic

Types of Architecture

Topic

Program Counter

Topic

Instruction Register

Topic

Stack Pointer

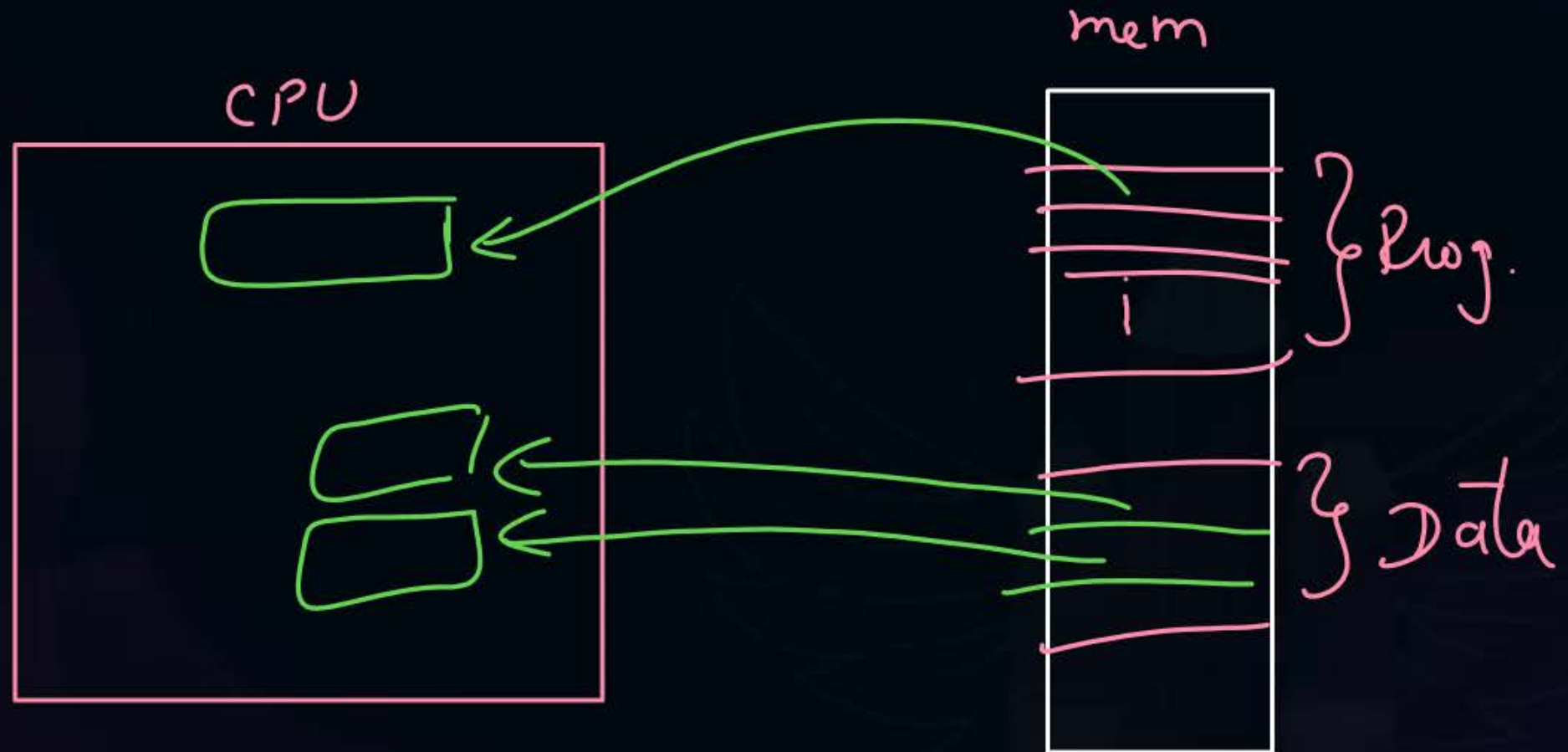


Topic : CPU Registers

small memories inside CPU to carry out program execution.



Reg.





Topic : CPU Registers



CPU Register

- General Purpose Registers (GPRs) $\Rightarrow R0, R1, R2, R3, R4, \dots$
- Special Purpose Registers (SPRs)



Topic : CPU Registers



CPU Register

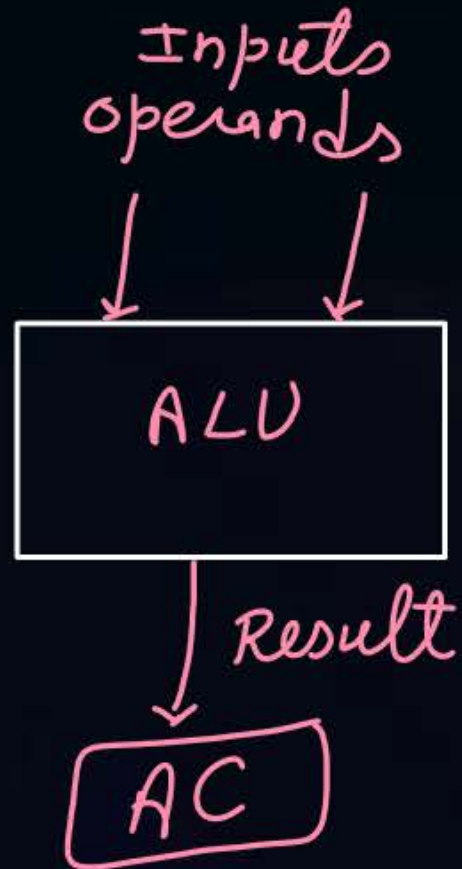
- General Purpose Registers (GPRs)
- Special Purpose Registers
 - ✓ 1. Accumulator (AC)
 2. Program Counter (PC)
 3. Instruction Register (IR)
 4. Stack Pointer (SP)
 5. Flag Register / Program Status Word (PSW) / *status Register*
 6. Address Register (AR) / Memory Address Register (MAR)
 7. Data Register (DR) / Memory Data Register (MDR) / MBR (*memory buffer Register*)



Topic : Accumulator

In AC-based architecture

- Used to store result of ALU and sometimes one of the operands for ALU too.





Topic : Types of Architecture

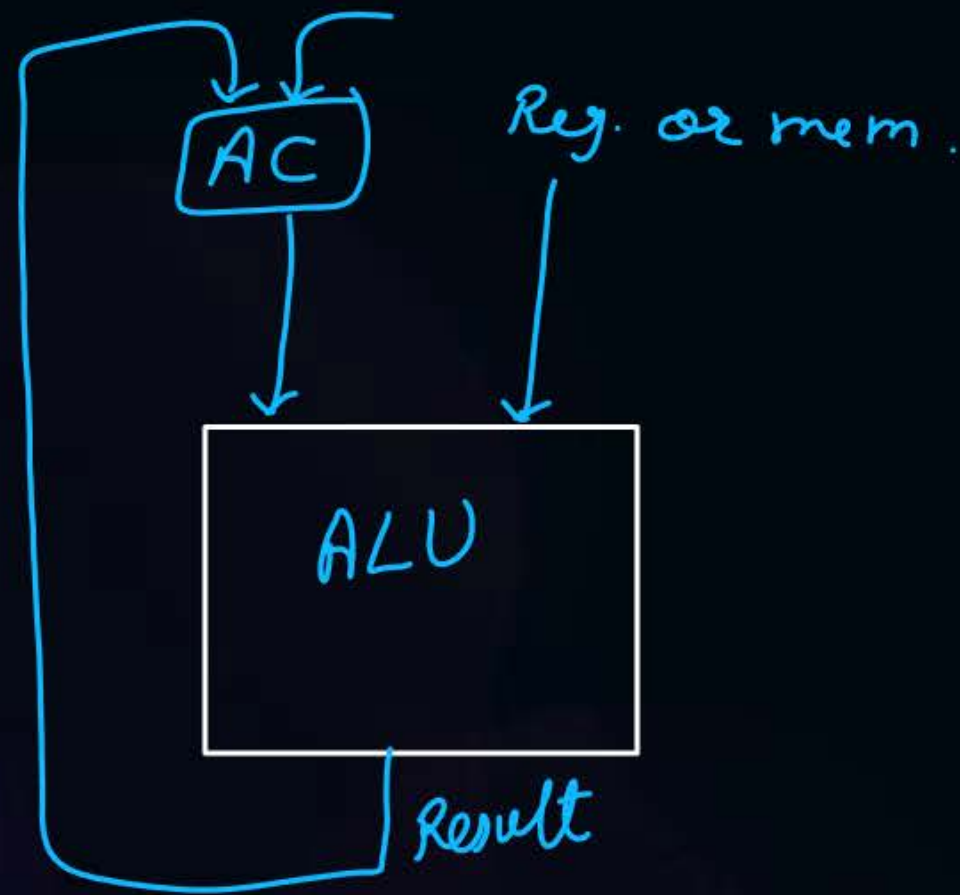
Based on ALU input: *from where input for ALU are taken*

- AC-Based Architecture
- Register Based Architecture
- Register-Memory Based Architecture
- Complex System Architecture
- Stack Based Architecture



Topic : AC-Based Architecture

one of the input operands should be taken from AC itself.
other one can be taken either from reg. or memory.



$A + B$

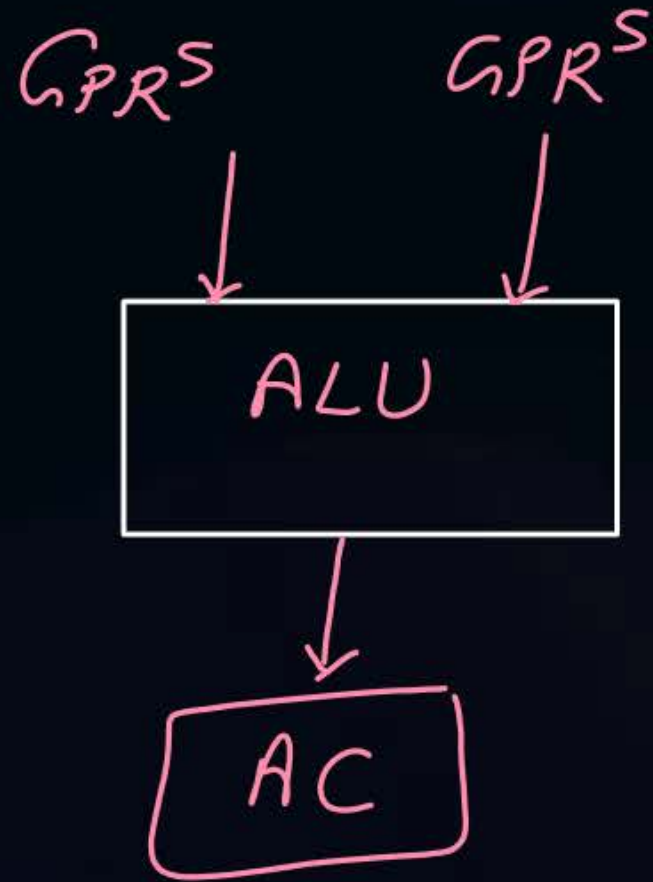
A, B are in mem.

$AC \leftarrow A$

$AC \leftarrow AC + B$



Topic : Register-Based Architecture



ex:-

$A + B$

A, B are memory

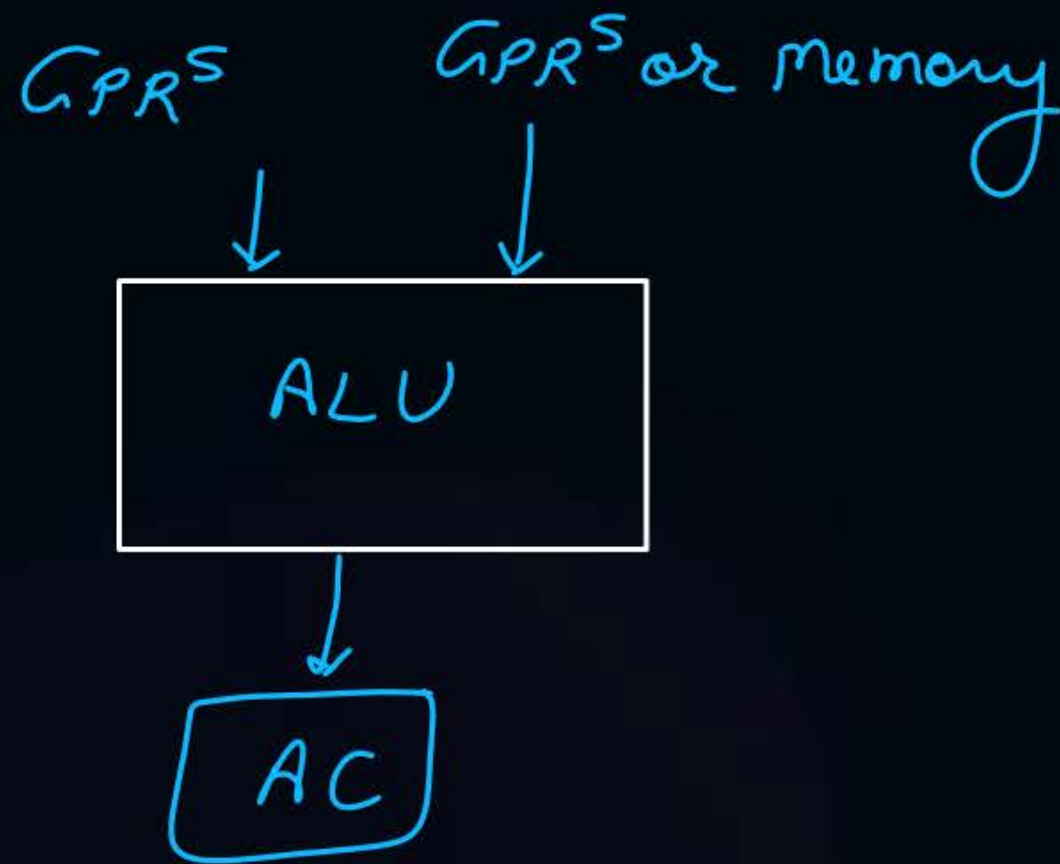
$R1 \leftarrow A$

$R2 \leftarrow B$

$AC \leftarrow R1 + R2$



Topic : Register-Memory Based Architecture



$A + B$

A, B are in memory

$R1 \leftarrow A$

$AC \leftarrow R1 + B$



Topic : Complex System Architecture

GPR^S or mem.

GPR^S or mem.



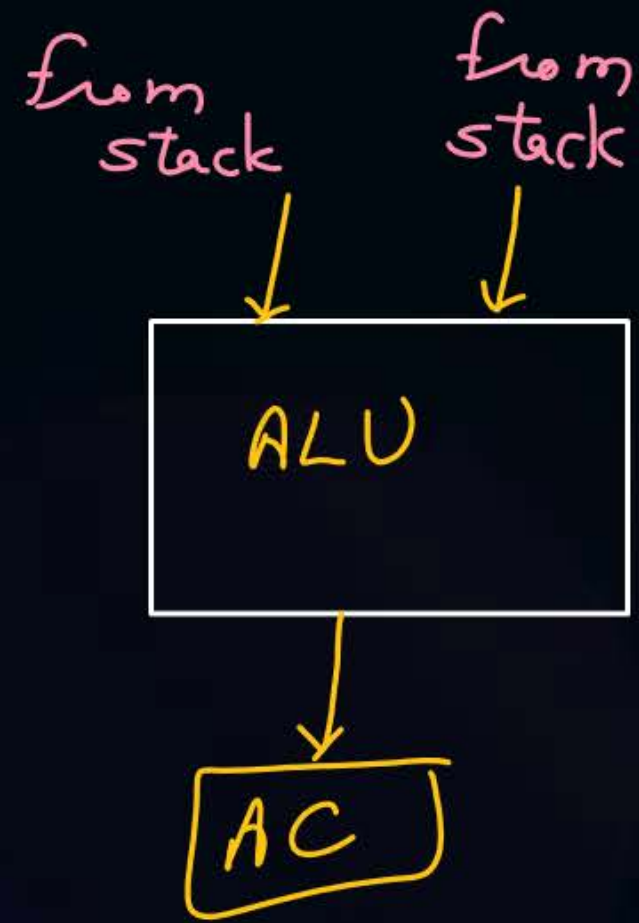
$A + B$

A, B are memory

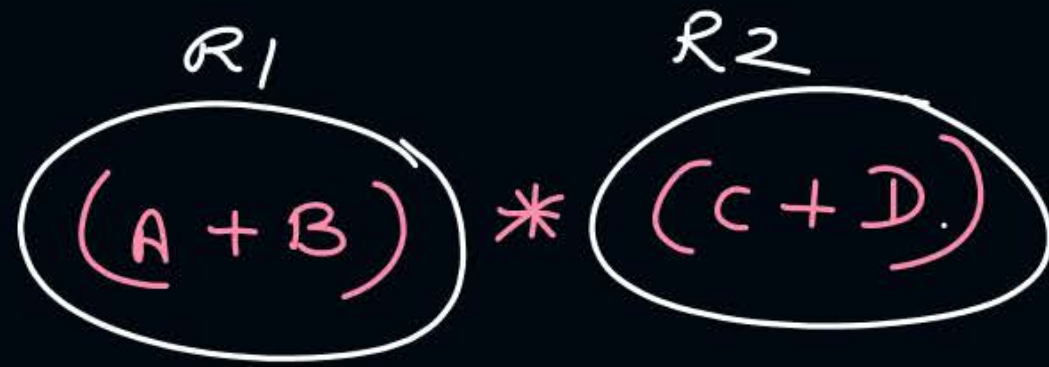
$$AC \leftarrow A + B$$



Topic : Stack-Based Architecture (outdated)



(stack in mem.)



A, B, C, D are mem.

Complex system arch. :-

$$AC \leftarrow A + B$$

$$R_1 \leftarrow AC$$

$$AC \leftarrow C + D$$

$$R_2 \leftarrow AC$$

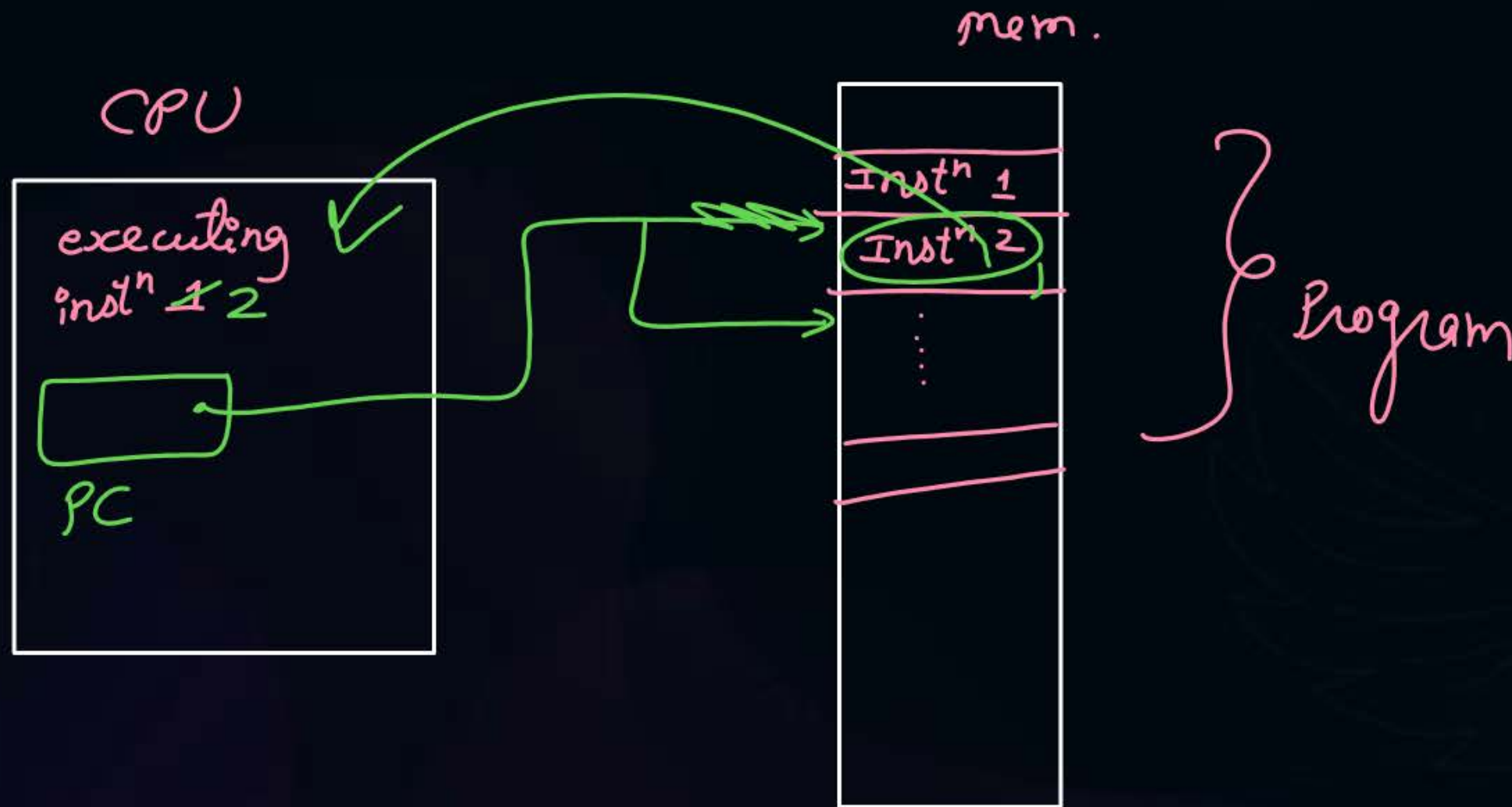
$$AC \leftarrow R_1 * R_2$$

AC - base arch :-



Topic : Program Counter

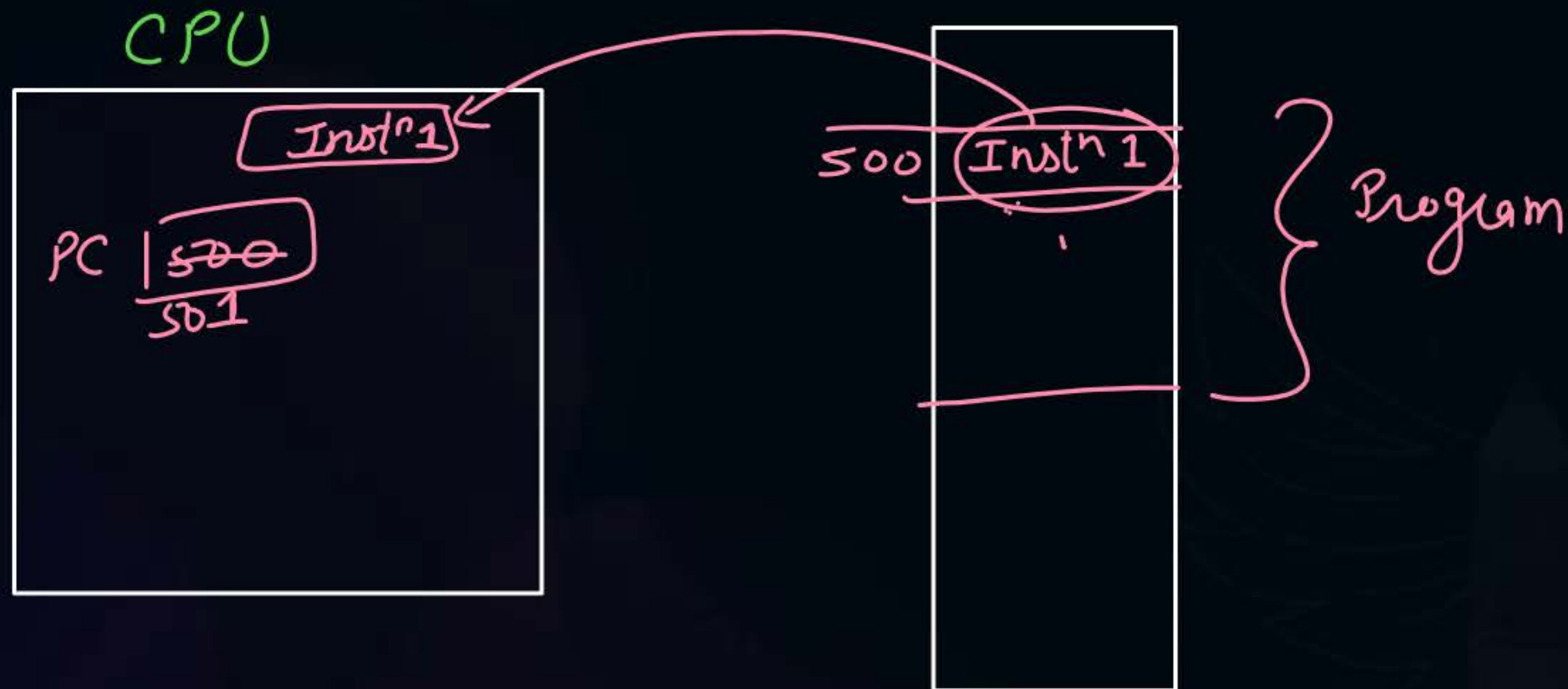
- Stores address of next instruction to be executed





Topic : Instruction Register

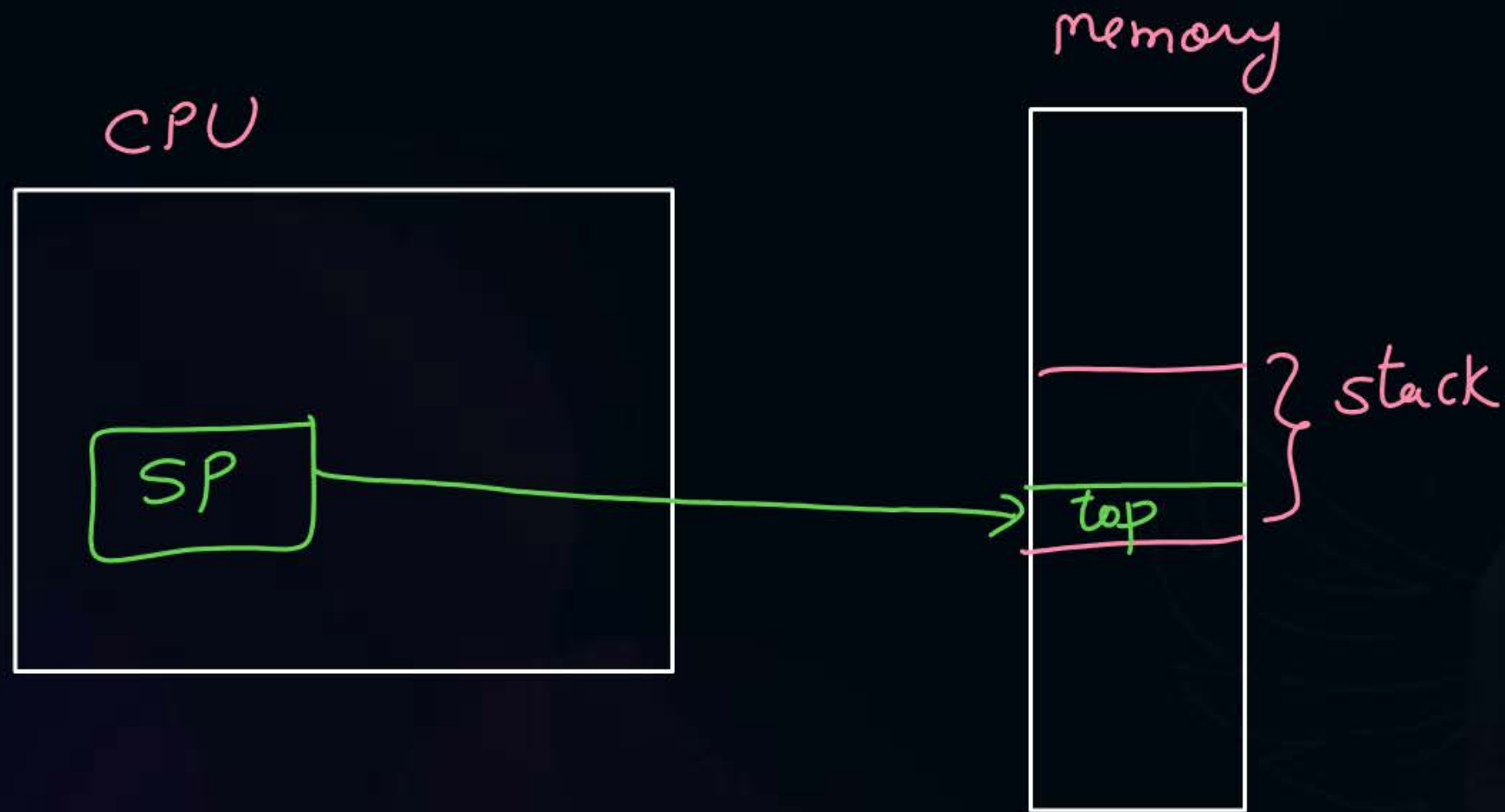
- Stores the current instruction to be executed





Topic : Stack Pointer

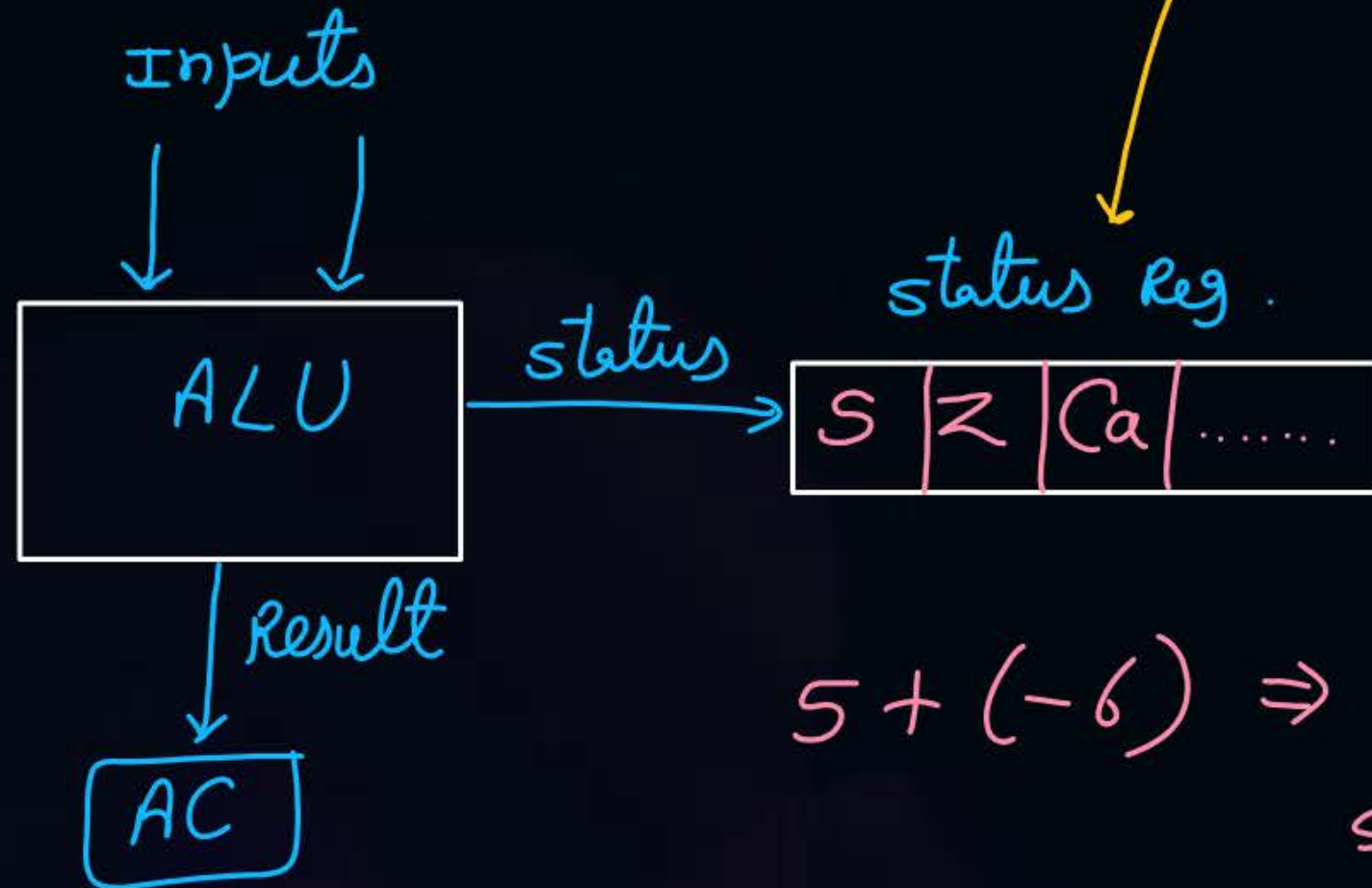
- Stores the address of the top of the stack





Topic : Flag or Status Register

- Stores the status of the ALU result



To implement conditions

S = Sign $\Rightarrow$ +ve or -ve

Z = Zero $\Rightarrow$ Zero or non-Zero

Ca = Carry $\Rightarrow$ Carry or not

$$5 + (-6) \Rightarrow \text{Result} = -1$$

status = -ve and non-Zero

ex:-

if ($a > b$)

$\Rightarrow$

performs $a - b$

check if result is +ve and non-zero
then condition is true



Topic : Address Register or MAR

- Used to send address to memory



Topic : Data Register or MDR



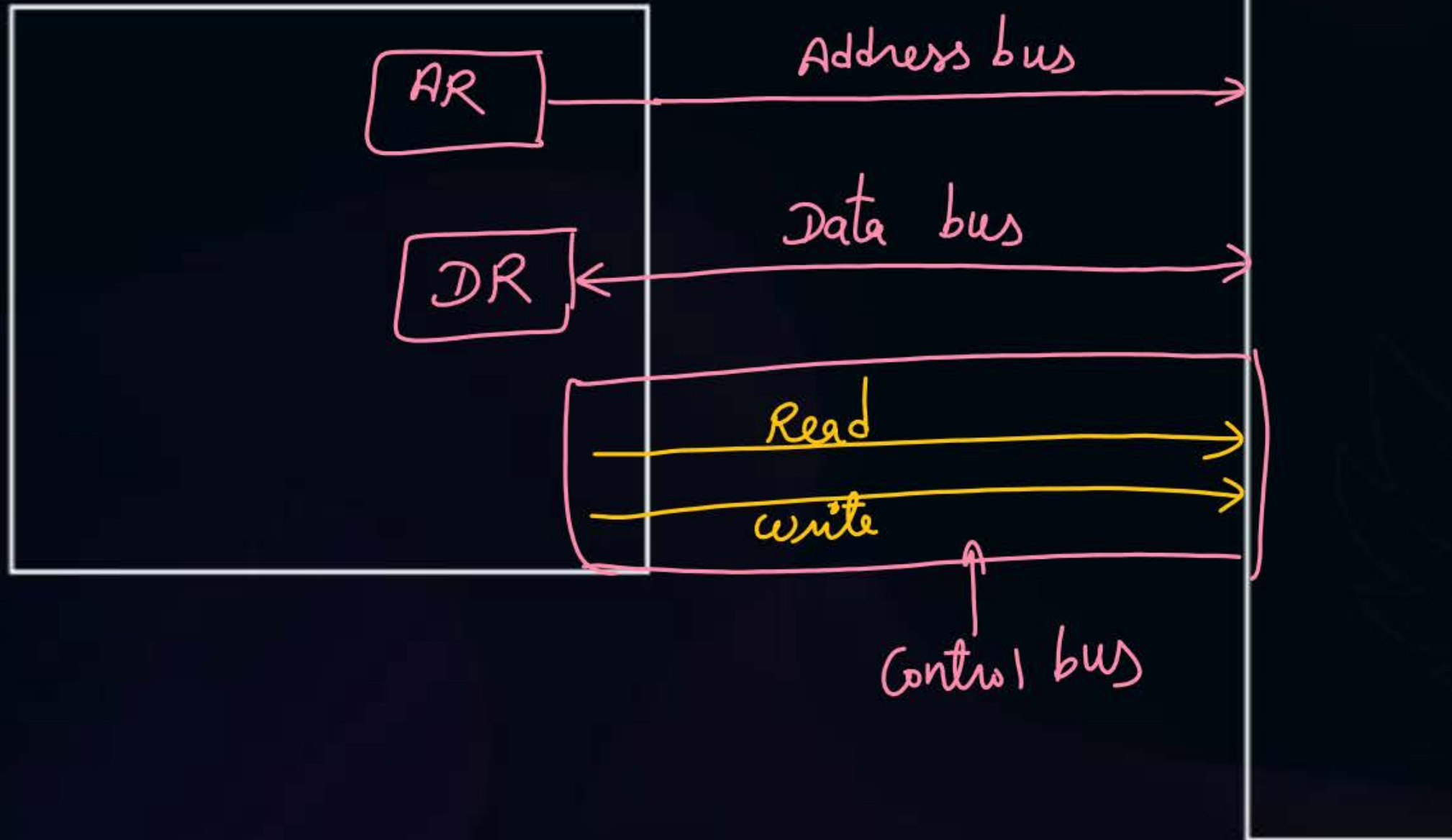
- Used to send data to memory (*for mem. write*)
- And to receive data from memory (*for mem. read*)



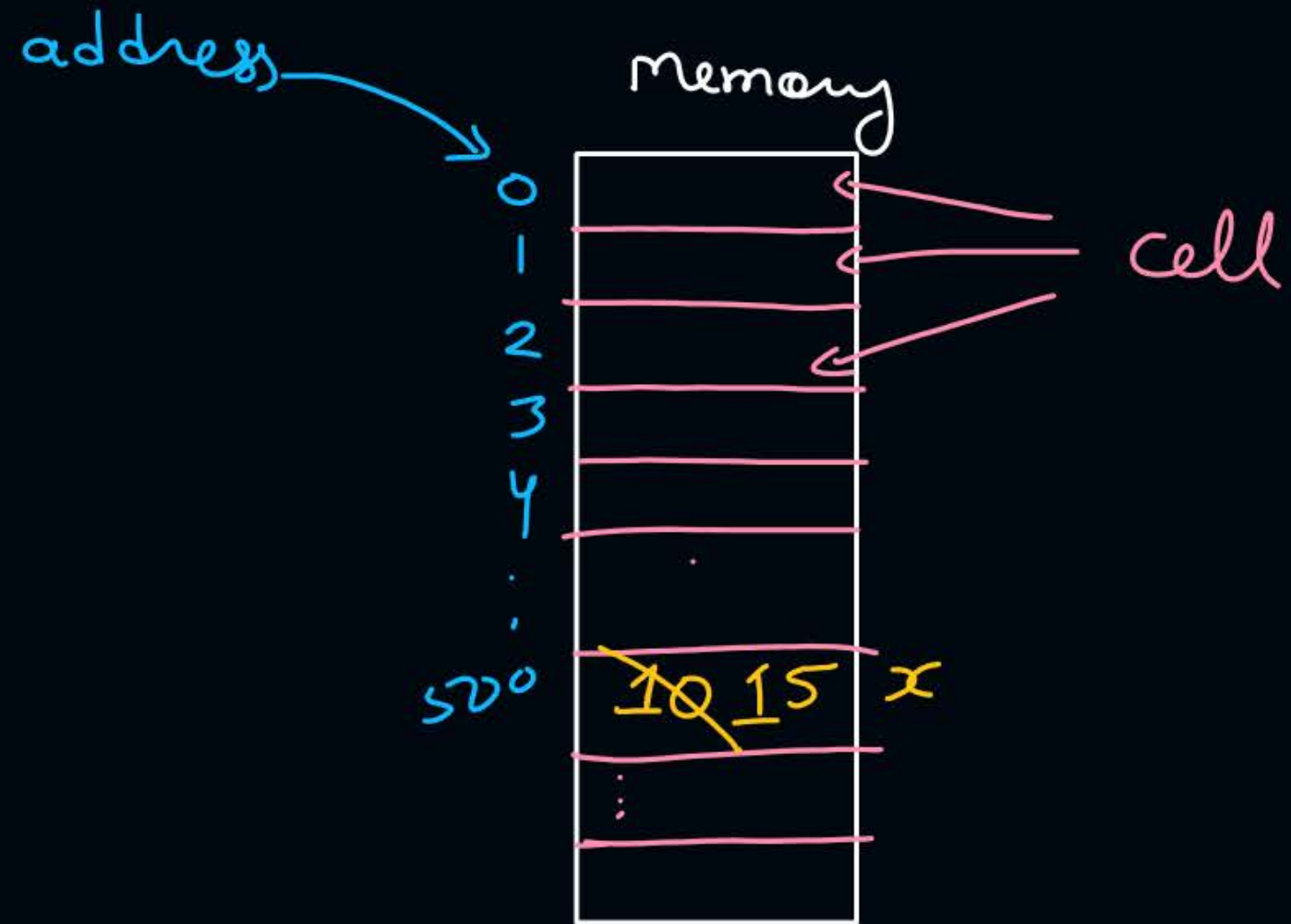
Topic : Memory Access

CPU

Memory



Memory :-



operations in memory :-

1. Read
2. write

Read :-

1. CPU sends address to mem. through add. bus
2. CPU sends active 'Read' control signal.
3. mem. performs read on given address and sends data to CPU through data.

write :-

1. CPU sends add. to mem. through add. bus
2.  data bus
3. CPU sends active 'write' control signal.
4. memory performs write of given data at given address.



2 mins Summary



Topic

CPU Registers

Topic

Types of Architecture

Topic

Program Counter

Topic

Instruction Register

Topic

Stack Pointer



Happy Learning

THANK - YOU